

Simulation of the movement of bentazon in soils using the CALF and PRZM models

P.H. Nicholls

To cite this article: P.H. Nicholls (1994) Simulation of the movement of bentazon in soils using the CALF and PRZM models, Journal of Environmental Science and Health . Part A: Environmental Science and Engineering and Toxicology, 29:6, 1157-1166, DOI: [10.1080/10934529409376100](https://doi.org/10.1080/10934529409376100)

To link to this article: <http://dx.doi.org/10.1080/10934529409376100>



Published online: 15 Dec 2008.



Submit your article to this journal [↗](#)



Article views: 6



View related articles [↗](#)



Citing articles: 4 View citing articles [↗](#)

Simulation of the Movement of Bentazon in Soils Using the CALF and PRZM Models

P.H. Nicholls

AFRC Institute of Arable Crops Research
Rothamsted Experimental Station
Harpenden, Hertfordshire, AL5 2JQ, U.K.

Abstract

Concentrations of bentazon in leachate draining from lysimeters containing intact monoliths of Nantuna sand were simulated using a model for leaching and degradation of pesticides in field soils (CALF) and the Pesticide Root Zone Model (PRZM). The CALF model empirically simulates the movement of water and the hydrodynamic dispersion of solutes by dividing the soil water into mobile and immobile categories; it Calculates Flow, hence the name, CALF. Sorption was calculated, using a linear isotherm, as a function of depth in soil. Bentazon degradation rates were calculated as functions of temperature, water content and depth in soil. Neither model simulated the early appearance of bentazon in leachate 100 days after application but both models closely estimated the cumulative amount of bentazon which had leached by day 325. Neither model made correct simulations when preferential flow of herbicide was thought to occur. The results of simulations by CALF and PRZM were similar.

Introduction

Trace concentrations of some pesticides reach drains and groundwater after being carried downward by water. Simulation of pesticide movement by such mass flow in water depends on the ability to simulate water movement accurately. Water flux is proportional to the difference in pore-water pressure between two points, the constant of proportionality being the hydraulic conductivity. Soil-water pressure and hydraulic conductivity are both sensitive functions of soil-water content. Most of the difficulties in simulating mass flow arise from dealing with pore-water pressure and hydraulic conductivity. The functions can be difficult to measure, and both functions, especially hydraulic conductivity, are notoriously and strongly variable. One practical way of side-stepping the use of hydraulic conductivity functions is by taking the capacity approach. In capacity models, certain ranges of soil-pore sizes are filled before others, and one of the most useful techniques is to allow the soil-water content to reach field capacity before draining. In many soils in temperate climates, water content remains fairly constant for long periods in winter due to the low evapotranspiration rates, so at those times drainage and, therefore, leaching is greatest.

In 1870, Lawes built some large brick lysimeters (area 0.001 acre; depth 1.0–1.5 m). In 1977, Addiscott et al. [1, 2] measured the leaching of inorganic nitrate and chloride through the structured soil in these lysimeters, and the results were simulated using a capacity model. The model assumed that soil-water content stayed at field capacity and that there were mobile and immobile categories of water with the division set at -200 kPa pressure. Incoming rainwater displaced water and solute in the mobile phase such that solute was simulated to appear in the leachate draining from the base of the lysimeter. Solute dispersion or spreading is also simulated because some of the solute remains associated with the immobile phase and does not move during that daily cycle. The results of the simulation of the leaching of nitrate and chloride ions were sufficiently encouraging to develop the model to include sorption and degradation of pesticides, and this model was used to simulate results of a leaching experiment using the herbicide fluometuron [3].

The simulation of leaching during winter alone was too restrictive, and the model (now called CALF) [4] was further developed to include soil drying by evaporation. The simulation of water movement by CALF is described in detail by Nicholls et al. [4]. The CALF model was designed to predict redistribution of pesticides largely in fallow soils down to plough depth. In this

respect, CALF has been useful in helping to understand problems with the efficacy of nematicides [5] and with herbicide injury to crops planted in subsequent seasons [6]. Usually, pesticides remain predominantly in the cultivated layer of soil until they are degraded, and models such as CALF [4] and the Pesticide Root Zone Model (PRZM) [7] have been useful in understanding such distributions. Thus, CALF was not originally designed to predict trace concentrations of solutes in leachate. One important feature of CALF is the small amount of input data required to do useful simulations. The minimum input data required by CALF might be:

- a. the rate of degradation measured for at least one well-defined value of soil-water content and air temperature;
- b. the linear coefficient of sorption to soil;
- c. the water content of soil at field capacity (-5 kPa); and
- d. daily weather data: rainfall, and maximum and minimum air temperatures.

Model Description

The CALF model utilizes soil layers and assigns each a thickness of 1 cm; it also uses weather data on a daily basis. Hydrological properties (field capacity and bulk density) were assumed constant with depth. Water flow across the boundary between two soil layers is calculated assuming the new water content to be the mean of both, whether simulating upward or downward movement. Only pesticides associated with mobile water (that between -5 and -200 kPa) are moved from one layer to another. For each layer after water and solute have been moved, pesticide concentrations in mobile and immobile soil phases are equalized and a new sorption equilibrium is established between the soil and soil solution.

Pesticide degradation is calculated as a function of soil temperature and water content using the routines of Walker and Barnes [8]. The sorption coefficient is calculated to increase gradually with time as has been observed in the field [9].

In the CALF model, any water which exceeds field capacity is immediately passed, together with its associated pesticide, to the layer below until it reaches the bottom of the profile. However, within this routine, when the water reaches the next layer, the pesticide is brought to sorption equilibrium before water is moved down once more. In this way, rapid flow is completed

during a 24-hour period and the water content of the soil profile is reset to the field capacity value.

The evaporation of water is calculated from maximum and minimum air temperatures using Linacre's method [10] as described by Walker and Barnes [8]. The evaporation of water from soil is calculated as a function of soil-water pressure using the method described by Nicholls et al. [4]. The simulations are for fallow soil.

The PRZM model is described by Mueller [11] (this issue); therefore, only limited reference to it is made here. This model was included because it is one of the most widely used models, but because the format for input data is different than for the CALF model, the two models cannot be compared using identical input data.

Lysimeter Experiments

The present paper gives some simulations of experimental data described in full by Bergström et al. [12]. They measured water drainage and bentazon leaching through lysimeters (depth 118 cm; diameter 29.5 cm) containing intact monoliths of Swedish soils.

Results

In laboratory incubation experiments dichlorprop was completely degraded within 2.2 or 10.9 days for Mellby and Lanna soils, respectively [12]. If such rapid rates of degradation are used as input data in CALF or PRZM models no dichlorprop is simulated to occur in measurable concentrations in the leachate.

Bentazon degrades more slowly than dichlorprop and so its movement in Nântuna sand was simulated using CALF and PRZM models. Input data used for the simulations are given in Tables 1–3. The sorption coefficient and degradation data were assumed values estimated from unpublished information. Levels of cumulative water drainage through Nântuna sand are shown in Figure 1. After 325 days, the drainage simulated by CALF was 263 mm and the mean measured value for three replicate lysimeters was 245 mm. This is considered sufficiently accurate for the simulation of pesticide leaching and is a good estimate from the use of easy-to-obtain input data. The input data were largely those measured and no calibration of, nor adjustments to, the model were done prior to the simulations of the movement of water.

Table 1: Input data used for CALF simulation of bentazon in Nântuna sand.

Item	Limits	Value
Field Capacity		0.168 kg/kg
Soil-water content at -200 kPa	(60% of field cap. value)	0.10 kg/kg
Bulk density		1.43 kg/dm ³
Linear sorption coeff. K_d	(0–20 cm)	0.2 dm ³ /kg
	(20–40 cm)	0.1
	(> 40 cm)	0.05
Half-life for degradation at 25 °C	(0–20 cm)	20 days at field cap.
	(20–40 cm)	40
	(> 40 cm)	80
Init. soil-water content		0.168 kg/kg
Rate of application		1.2 kg/ha
Latitude of site		59.5° N
Altitude of site		20 m
Daily: rainfall, maximum and minimum air temperature		

Table 2: Input data used in PRZM model simulations of bentazon in Nântuna sand.

Depth (cm)	Bulk density (kg/dm ³)	Wilting point (m ³ /m ³)	Field cap. (m ³ /m ³)	Initial water content (m ³ /m ³)	Half-life (day)	K_d (dm ³ /kg)
0–20	1.43	0.034	0.236	0.236	20	0.9
20–85	1.43	0.034	0.236	0.236	40	0.26
85–105	1.46	0.022	0.115	0.115	150	0.2
Water solubility					350 mg/L	
Application rate					1.2 kg/ha	

Table 3: Input data used in PRZM model simulations of bentazon in Nântuna sand.

Parameter	Value
Pan factor	0.60
Snow factor	0.45
Pan evaporation flag	0
Min. depth evaporation extracted	15 cm
Initial crop number	1
Erosion flag	0
Number of crop simulations	1
Crop number	1
Max. interception crop storage	0.01 cm
Max. active crop root depth	40 cm
Max. areal crop coverage	50%
Number cropping periods	1
Number of pesticide applications	1
Depth of pesticide incorporation	0 cm
Pesticide application model	1
Soil core depth (total)	105 cm
Plant uptake efficiency factor	0
Total no. of simulation compartments	21
Bulk density flag	0
Soil moisture calculation flag	0
Pesticide sorption calc. flag	0
Switch for soil hydraulics	0

Both PRZM and CALF predicted too little evaporation of water in the first 100 days because the presence of a crop was not simulated. Neither model was set to simulate the freezing conditions in December 1989 and January 1990, so drainage was simulated to continue during this period.

Bentazon levels in the leachate are shown in Figure 2. The cumulative bentazon level in leachate after 325 days was simulated by CALF to be 0.171 mg/m^2 and the mean measured value was 0.158 mg/m^2 . Neither model predicted the observed early breakthrough of bentazon. Results of simulations were sensitive to values of sorption and rates of degradation chosen for the soil below the cultivated layer. It may have been possible to improve the

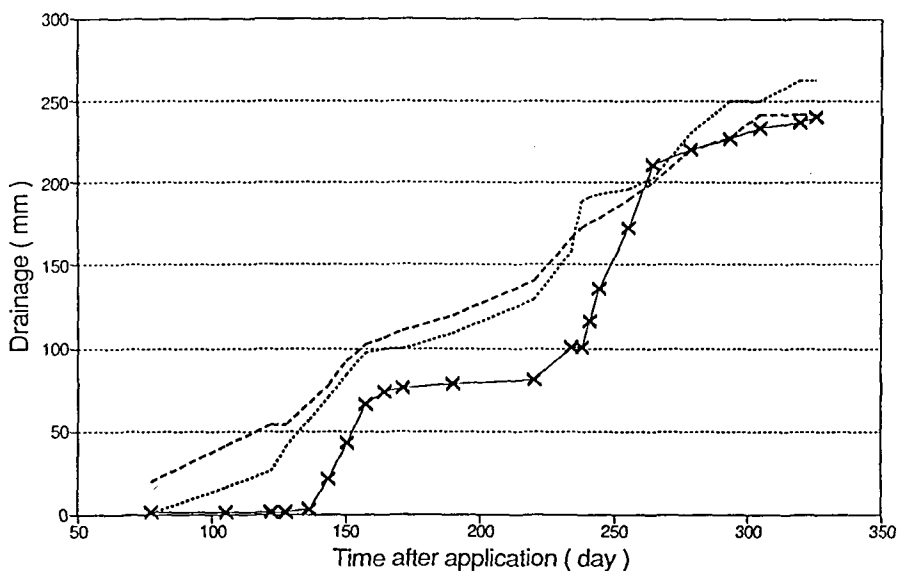


Figure 1: Measurements and simulations of water drainage from lysimeters of Nântuna sand. Measurements represent the mean of values from three lysimeters. Measured values —X—; CALF simulation ···; PRZM simulation ----.

fit between measured and simulated values by finely adjusting such input data but in the absence of measured data this would not have been sufficiently informative to be justifiable. However, the sorption and degradation half-life values used in PRZM were somewhat greater than those used with CALF to give similar concentrations in the leachate (see Tables 1–3).

Simulations were made of the leaching of bentazon in Mellby and Lanna soils using input that was as close as possible to that given by Bergström et al. [12]. Bentazon was predicted not to occur in leachate from the lysimeters, which agrees with the observation from Mellby sand while only a trace of bentazon was observed in leachate from Lanna clay. The simulations were made without prior knowledge of the observed water drainage or concentrations of bentazon in leachate.

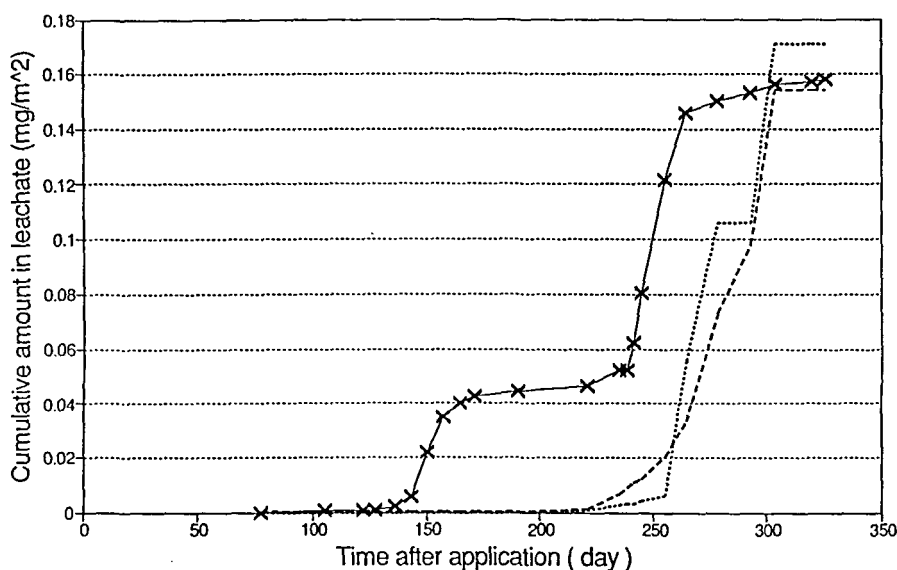


Figure 2: Measurements and simulations of cumulative amounts of bentazon in leachate from lysimeters of Nântuna sand. Measured values —X—; CALF simulation ···; PRZM simulation ---.

Conclusions

The CALF model closely simulated drainage and the final cumulative amount of bentazon in leachate from lysimeters containing Nântuna sand. Simulation results from CALF were similar to those obtained using PRZM model. Nevertheless, the prediction of pesticide concentrations in leachate is unsatisfactory unless models are simultaneously able to simulate accurately the distributions of the pesticide throughout the soil profile. Measured distributions of the pesticide in the soil profile were not given for these experiments, so comparisons with simulations could not be made.

Although CALF uses the concept of mobile and immobile water categories, the movement of a pesticide with preferential and bypass flow is only simulated to occur downward for 1 cm before the pesticide reaches sorption

equilibrium for any given depth. Hence, deep penetration of pesticide by preferential flow is not simulated, and this is an additional reason why no dichlorprop was simulated to occur in the leachate from the lysimeter experiments.

In the future, CALF will be developed to incorporate preferential and bypass flow mechanisms and will be tested with measured data from laboratory, lysimeter and field experiments.

Acknowledgment

The author is grateful for computations and simulations made by Mr. A. Nacev.

Notation

CALF a model for leaching and degradation of pesticides in field soils that

CALculates Flow

PRZM Pesticide Root Zone Model

References

- [1] T.M. Addiscott. A simple computer model for leaching in structured soils. *J. Soil Sci.*, 28:554–563, 1977.
- [2] T.M. Addiscott, D.A. Rose, and J. Bolton. Chloride leaching in Rothamsted drain gauges. *J. Soil Sci.*, 29:305–314, 1978.
- [3] P.H. Nicholls and T.M. Addiscott. Computer simulation of herbicide leaching in a structured soil. In *Proc. British Crop Protection Conference - Weeds*, pages 617–620. 1980.
- [4] P.H. Nicholls, A. Walker, and R.J. Baker. Measurement and simulation of the degradation of atrazine and metribuzin in a fallow soil. *Pestic. Sci.*, 12:484–494, 1982.
- [5] D.A. Cooke, R.H. Bromilow, and P.H. Nicholls. The extent and efficacy of granular pesticide usage to control ectoparasitic nematodes on sugar beet. *Crop Protec.*, 4:444–457, 1985.

- [6] P.H. Nicholls, A. Walker, and A.A. Evans. The behaviour of chlorsulfuron and metsulfuron in soils in relation to incidents of injury to sugar beet. In *Proc. 1987 British Crop Protection Conference - Weeds*, pages 549–556. 1987.
- [7] R.F. Carsel, L.A. Mulkey, M.N. Lorber, and L.B. Baskin. The pesticide root zone model (PRZM): A procedure for evaluating pesticide leaching threats to groundwater. *Ecol. Model.*, 30:46–69, 1985.
- [8] A. Walker and A. Barnes. Simulation of herbicide persistence in soils; a revised computer model. *Pestic. Sci.*, 12:123–132, 1981.
- [9] A. Walker. Evaluation of a simulation model for prediction of herbicide movement and persistence in soil. *Weed Res.*, 27:143–152, 1987.
- [10] E.T. Linacre. A simple formula for estimating evaporation rates in various climates using temperature data alone. *Agricultural Meteorology*, 23:357–363, 1977.
- [11] T.C. Mueller. Comparison of PRZM computer model predictions with field lysimeter data for dichlorprop and bentazon leaching. *J. Environ. Sci. Health*, 1994. This issue.
- [12] L. Bergström, N. Jarvis, and J. Stenström. Pesticide leaching data to validate simulation models for registration purposes. *J. Environ. Sci. Health*, 1994. This issue.

Received: July 22, 1993

Accepted: August 30, 1993